

JEE MAIN REVISION MASTER: DIFFERENTIAL EQUATIONS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 244 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Variable Separable & Homogeneous DE	213 PYQs	Separable DE
Very High	Linear Differential Equations & Integrating Factor	49 PYQs	Linear DE

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Variable Separable & Homogeneous DE	$\int \frac{dy}{g(y)} = \int f(x) dx + C$, $\quad \text{For homogeneous: substitute } y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$
Linear Differential Equations & Integrating Factor	$\frac{dy}{dx} + P(x)y = Q(x) \implies \text{I.F.} = e^{\int P(x) dx}$, $\quad y \cdot \text{I.F.} = \int Q(x) \cdot \text{I.F.} dx + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Variable Separable & Homogeneous DE (Very High Yield — 213 PYQs)

Core Principle: Classification of order and degree, separation of variables, and reduction of homogeneous equations via $y = vx$.

■ **Caution / JEE Pitfall:** Degree of a differential equation is defined ONLY when the equation is a polynomial in its derivatives (no fractional powers or $\sin(dy/dx)$).

■ **Speed Trick:** Equations of the form $\frac{dy}{dx} = f(ax + by + c)$ can be solved by substituting $z = ax + by + c$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy + (y\sqrt{1-x^2} - x\cos^{-1}x)dx = 0, x \in (0,1)$, $\underset{x \rightarrow 1}{\lim} y = ?$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $(\frac{dy}{dx} + 1 + x + x^2)(1 - y + y^2) = 0, y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let $y=y(x)$ be the solution of the differential equation $x(\frac{dy}{dx} - y) = x^2 \cot x, x \in (0,\pi)$. If $y(\frac{\pi}{2}) = \frac{\pi}{2}$..." [Ans: N/A]

3.2 Linear Differential Equations & Integrating Factor (Very High Yield — 49 PYQs)

Core Principle: Standard first-order linear differential equations and integrating factor technique for $y(x)$ and $x(y)$.

■ ■ **Caution / JEE Pitfall:** If the equation has $x \frac{dy}{dx}$, remember to divide by x throughout first so the coefficient of $\frac{dy}{dx}$ is 1!

■ **Speed Trick:** If the equation is non-linear in y , check if it is linear in x : $(dx/dy) + P(y)x = Q(y)$ implies I.F. = $e^{\int P(y)dy}$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Let $y=y(x)$ be the solution of the differential equation $x(dy/dx)-y=x^2\cot x$, $x \in (0,\pi)$. If $y(\frac{\pi}{2})=\frac{\pi}{2}$..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Let $f:R \rightarrow R$ be such that $f(xy)=f(x)f(y)$, for all $x,y \in R$ and $f(0) \neq 0$. Let $g:[1,\infty) \rightarrow R$ be a differe..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#2 (MCQ):** "Let $y=y(x)$ be the solution of the differential equation $x(dy/dx)-\sin 2y=x^3(2-x^3)\cos^2 y$, $x \neq 0$. If $y(2)=0$, then $\tan(y(1))=$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let $y=y(x)$ be the solution of the differential equation $(dy / dx)=(1+x+x^2)(1-y+y^2)$, $y(0)=(1 / 2)$. Then $(2y(1)-1)$ is equal to

- (A) $\sqrt[3]{\tan\left(\frac{11\sqrt{3}}{6}\right)}$
- (B) $\frac{1}{\sqrt[3]{2\tan\left(\frac{11\sqrt{3}}{12}\right)}}$
- (C) $\sqrt[3]{\tan\left(\frac{11\sqrt{3}}{12}\right)}$ **[Correct]**
- (D) $\frac{1}{\sqrt[3]{2\tan\left(\frac{11\sqrt{3}}{6}\right)}}$

Correct Answer: C

Solution Analysis: Given $(dy / dx)=(1+x+x^2)(1-y+y^2)$, $y(0)=(1 / 2)$ We will solve it by the method of separation of variables. Since $\frac{dy}{1-y+y^2}=(1+x+x^2)dx$ Now simplify the denominator on the left: $1-y+y^2=y^2-y+1=(y-(1 / 2))^2+(3 / 4)$ So, $\int \frac{dy}{(y-(1 / 2))^2+(\frac{\sqrt{3}}{2})^2}=\int (1+x+x^2)dx$ Using the standard result $\int \frac{du}{u^2+a^2}=(1 / a)\tan^{-1}(u / a)$ we get $\frac{2}{\sqrt{3}}\tan^{-1}(\frac{2y-1}{\sqrt{3}})=x+\frac{x^2}{2}+\frac{x^3}{3}+C$ Now use the initial condition $y(0)=(1 / 2)$. Then $2y-1=0$ so $\$)$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#3)

Let $f: [1, \infty) \rightarrow \mathbb{R}$ be a differentiable function defined as $f(x) = \int_1^x (1)^t f(t) dt + (1-x)(\log(e)x - 1) + e$. Then the value of $f(f(1))$ is :

- (A) $(1+e^e)$ **[Correct]**
- (B) $(1+e)$
- (C) $(1+e+e^e)$
- (D) $1+2e$

Correct Answer: A

Solution Analysis: Given $f(x) = \int_1^x (1)^{x^t} dt + (1-x)(\ln x - 1) + e$, $x \in [1, \infty)$. We have to find $f(f(1))$. First, put $x=1$ in the given equation: $f(1) = \int_1^1 (1)^{x^t} dt + (1-1)(\ln 1 - 1) + e$. Now, $\int_1^1 (1)^{x^t} dt = 0$, $(1-1)(\ln 1 - 1) = 0$. So, $f(1) = e$. Hence, $f(f(1)) = f(e)$. Now we find $f(x)$. Differentiate both sides of $f(x) = \int_1^x (1)^{x^t} dt + (1-x)(\ln x - 1) + e$. Using the Fundamental Theorem of Calculus, $\frac{d}{dx} \left(\int_1^x (1)^{x^t} dt \right) = f(x)$. So, $f'(x) = f(x) + \frac{d}{dx} [(1-x)(\ln x - 1)]$. Now differentiate the second term carefully: $\frac{d}{dx} [(1-x)(\ln x - 1)] = -(\ln x - 1) + (1-x) \cdot \left(\frac{1}{x} \right)$. Simplifying, $f' = -\ln x + 1 + \frac{1-x}{x}$.